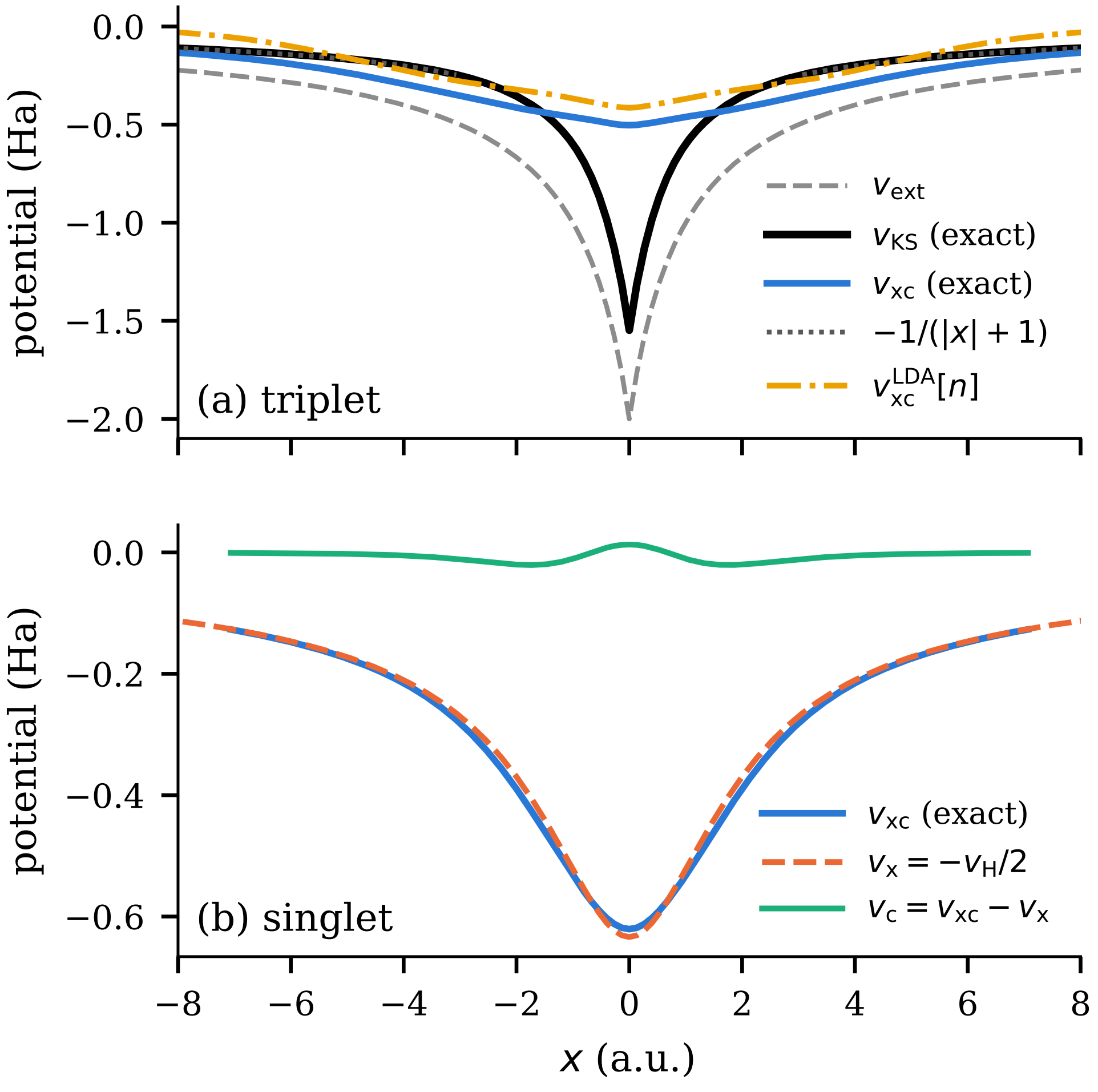
potential (Ha)
0.0
−0.5
−1.0
−1.5
−2.0
(a) triplet
v_{ext}
v_{KS} (exact)
v_{xc} (exact)
$-1/(|x|+1)$
$v_{\mathrm{xc}}^{\mathrm{LDA}}[n]$
potential (Ha)
0.0
−0.2
−0.4
−0.6
(b) singlet
v_{xc} (exact)
$v_{\mathrm{x}} = -v_{\mathrm{H}}/2$
$v_{\mathrm{c}} = v_{\mathrm{xc}} - v_{\mathrm{x}}$
−8
−6
−4
−2
0
2
4
6
8
x (a.u.)